CO 2 - ASSESSMENT TOOL 1
Analytical Problem Solving

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Course: Cryptography and Network Security

1. RSA Encryption Analysis

Given:

$p = 11$

$q = 13$

$e = 7$

$M = 9$

a. Compute n and $\phi(n)$

$n = 11 \times 13 = 143$

$\phi(n) = (11-1)(13-1) = 120$

b. Private Key

$d = 103$ (since $7 \times 103 \equiv 1 \pmod{120}$)

Private Key = (103, 143)

c. Encryption

$C = 9^7 \bmod 143 = 48$

d. Decryption

$48^{103} \bmod 143 = 9$

e Effect of changing e

Smaller e: Faster encryption, lower security if poorly chosen.

Larger e : Slower encryption, more secure

Common value: 65537.

2. Block Cipher Mode Evaluation

Given:

Plaintext = [1010, 1010, 1111, 1010]

Key = 1100

IV = 0000

a. Encryption

ECB

1010 -> 0110

1010 -> 0110

1111 -> 0011

1010 -> 0110

Ciphertext : [0110, 0110, 0011, 0110]

CBC

C1 -> 0110

C2 = 0000

C3 = 0011

C4 = 0101

Ciphertext: [0110, 0000, 0011, 0101]

b. Comparison

ECB : Repeated ciphertext blocks

CBC : Different ciphertext blocks

c. Conclusion

CBC is better because it hides Plaintext patterns.

3. DES vs 3-DES vs Blowfish

(i) Total Time (1000 Blocks)

ALGORITHM	TIME
DES	28
3-DES	68
Blowfish	18

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DES: Fast but weak security
3-DES: Stronger but slow
Blowfish: Strong and fastest

iii Recommendation

Blowfish - Best for secure real-time communication

4. Diffie - Hellman Key Exchange

given:

$P = 23, g = 5$

$A = 6, B = 15$

d. Public Keys

$A = 5^6 \bmod 23 = 8$

$B = 5^{15} \bmod 23 = 19$

e Shared Secret

$19^6 \bmod 23 = 8^{15} \bmod 23 = 2$

f. MITM Attack

Attacker intercepts and replaces public keys
Creates separate shared keys with A & B

g. Prevention

Use Digital Signatures, Certificates, or

Authenticated Diffie-Hellmon (TLS).

5. ECC vs RSA

Given:

RSA = 1024 bits

ECC = 160 bits

IoT Device

d. Computational cost

RSA : high

ECC : low

e. Power and Memory

RSA: More Power and Memory

ECC: Less Power and Memory

f. Recommendation

ECC is best because it:

Uses smaller keys

Is faster

Consumes less power and memory

Is ideal for IoT devices.